ZEBRAFISH KNOWLEDGE SERIES

The Role of Zebrafish in Science

A small freshwater fish and the large part it plays in understanding life

An introduction to the zebrafish as a research animal: where it comes from, why it is easy to study, how closely its genes relate to ours, and where the limits of the model lie.

NZINC Nepal Zebrafish in Nature Center

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1 Overview

The zebrafish is a freshwater fish about the length of a paperclip. It is also one of the most widely used animals in biological research.

Danio rerio is native to South Asia, with wild populations in India, Pakistan, Bangladesh, Nepal and Bhutan, mainly in the Ganges and Brahmaputra basins. Wild adults are small, typically 1.8 to 3.7 cm long, and live in shallow, slow-moving water such as streams, ponds, ditches and rice paddies.

It entered the laboratory in the 1970s and 1980s, when George Streisinger and colleagues at the University of Oregon developed it as a genetic model. Their zebrafish clones were among the earliest successful vertebrate clones. Since then the species has become a standard model for studying how vertebrates develop, how genes function and how diseases begin.

2 Why researchers use it

Most of the zebrafish's value as a research animal comes from practical biology. The embryo develops outside the mother and is transparent, so cell movements, organ formation and the effect of a gene or drug can be watched in a living animal under a microscope.

TRAIT	WHAT IT MEANS FOR RESEARCH
Adult size	Wild adults are 1.8 to 3.7 cm. Large numbers can be kept in small spaces at low cost.
Embryo	Develops outside the mother. Fertilised eggs become transparent almost immediately.
Speed of development	Precursors of all the major organs are present within 36 hours of fertilisation.
Generation time	About three months, which allows genetic crosses over several generations in a year.
Egg numbers	One pair can lay 200 to 300 eggs in a single morning.
Lifespan	2 to 3 years in captivity. Wild fish are typically annual.

Sources: Animal Diversity Web; Wikipedia, "Zebrafish" (summary of primary literature).

3 Genetic similarity to humans

In 2013 the Wellcome Trust Sanger Institute published the zebrafish reference genome in *Nature* (Howe et al.). The genome is about 1.4 billion base pairs long and contains more than 26,000 protein-coding genes. Comparing it with the human genome showed a close relationship at the level of individual genes.

Figure 1. Gene-level similarity between humans and zebrafish. Source: Howe et al., *Nature*, 2013, as reported by the Sanger Institute.

The 84 per cent figure is the one that matters for medicine. For most human disease genes, a zebrafish counterpart exists that can be altered, switched off or studied in the animal. Fish went through an additional round of genome duplication in their evolutionary history, so a single human gene sometimes corresponds to two zebrafish genes. This complicates some experiments and is useful in others.

4 Tools built around the model

- **Forward genetic screens.** Large screens in Tübingen and Boston, in which researchers look for mutants with an altered body plan or organ, established the species as a leading model for vertebrate development.
- **Genome editing.** CRISPR-Cas9 protocols are well established, so a chosen gene can be disrupted or altered directly in the embryo.
- **Transgenic lines.** Fluorescent markers let researchers follow single cell types, such as blood vessels or immune cells, in the living animal.
- **Transparent adults.** The *casper* strain (2008) stays see-through into adulthood, allowing tumours and transplanted cells to be followed without dissection.

5 Limits of the model

A fish is not a mammal. Zebrafish have no lungs or mammary glands, and their physiology differs from ours in ways that matter for some diseases. Results from zebrafish are most useful as an early, fast and inexpensive step, and findings that bear on human health are normally confirmed in mammalian systems before they are applied.

6 Laboratory fish and wild fish

Almost all research uses a handful of inbred laboratory strains that have been bred for many generations. A population genomic study of wild and laboratory zebrafish (Whiteley et al., 2011) sampled 13 wild populations from Nepal, India and Bangladesh and compared them with three common laboratory strains. It found that the laboratory strains were genetically distinct from every wild population, and concluded that a large share of the variation present in the species is missing from the laboratory stocks.

Wild populations are therefore a genetic reserve for the model organism, in addition to being part of the ecosystems where they live. The companion papers in this series look at the impact of zebrafish research, the case for conserving wild zebrafish and their habitat, and the position of Nepal within the species' natural range.

Key points

- The zebrafish is native to South Asia, including Nepal.
- About 70% of human protein-coding genes, and 84% of human disease-associated genes, have a zebrafish counterpart.
- Transparent embryos, fast development and large clutches make it practical to study at scale.
- Laboratory strains hold only part of the genetic variation found in wild populations.

Sources

Howe K. et al. (2013). The zebrafish reference genome sequence and its relationship to the human genome. *Nature* 496: 498–503. Summary: Wellcome Trust Sanger Institute, via ScienceDaily, 17 April 2013.

Whiteley A.R. et al. (2011). Population genomics of wild and laboratory zebrafish (*Danio rerio*). *Molecular Ecology* 20(20): 4259–4276.

Animal Diversity Web, University of Michigan Museum of Zoology: *Danio rerio* (animaldiversity.org/accounts/Danio_rerio).

Zebrafish. Wikipedia (en.wikipedia.org/wiki/Zebrafish), used as a summary of the primary literature on embryo development, strains and tools.